Project 7 - Multi Agent RL - Phase 1

Code Repository Link: Multi Agent RL - Phase 1.

We use **BenchMARL** as the experimental framework for this phase. BenchMARL is a multi-agent reinforcement learning benchmark library built on top of TorchRL, with a strong emphasis on modular configuration, reproducibility, and fair comparison across algorithms and environments. BenchMARL organizes experiments through reusable YAML configurations and a unified training and evaluation pipeline, enabling consistent and reproducible comparisons across settings.

Algorithms

We study two PPO-based multi-agent baselines: **IPPO** and **MAPPO**. Both are actor–critic methods derived from PPO, but they differ in the information available to the critic during training.

IPPO (Independent PPO) models each agent as an independent learner with a decentralized policy and a critic based on local information. This makes IPPO conceptually simple and scalable, but it may be less effective in environments requiring strong coordination.

MAPPO (Multi-Agent PPO) follows the centralized-training, decentralized-execution paradigm. During execution, each agent acts using only its own observation, but during training the critic can access richer global information about the environment and other agents. This additional information can improve value estimation and coordination, making MAPPO a stronger baseline in settings where joint behavior is important.

This contrast allows us to directly evaluate the impact of centralized value estimation on multi-agent coordination.

Tasks

We evaluate the algorithms on four tasks spanning two environment families: **VMAS** and **PettingZoo**.

1. VMAS Navigation: The goal is for agents to move efficiently to their targets without interfering with one another. Formally, this is a cooperative continuous-control task with 3 agents evolving in a shared 2D environment. In our setup, collisions are enabled and each episode has a fixed horizon of 100 steps. The reward reflects coordinated progress toward task completion, while the interaction dynamics require agents to account for the presence of others in a shared space. Performance depends on both individual control and multi-agent coordination.

2. VMAS Balance: The goal is for the team to jointly stabilize and transport a shared object. This task involves 4 agents and a horizon of 100 steps. Unlike purely independent navigation, the environment couples the agents through a common physical objective, so the effect of one agent's action depends strongly on the actions of the others. The reward therefore reflects collective control quality, making this task a useful setting for evaluating whether centralized value estimation improves coordinated behavior.

3. PettingZoo Multiwalker: The goal is to carry the package forward without the walkers collapsing. This is a cooperative locomotion task with 3 walkers and a maximum episode length of 500 steps. The reward includes forward-progress incentives together with penalties for instability: in our configuration, a fall incurs a penalty of `-10`, and if a fall triggers episode termination, an additional termination penalty of `-100` is applied. Parameters like terminate_on_fall & remove_on_fall are set to True, which makes failure events significant both in return and in episode length. Consequently, strong performance requires both sustained forward motion and stable coordinated locomotion.

4. PettingZoo Simple Tag: This is a competitive multi-agent environment where a group of agents attempts to tag an adversarial agent. The setting involves asymmetric roles and partially conflicting objectives, making coordination more challenging than in fully cooperative tasks. Performance depends on both effective pursuit strategies and robustness to adversarial behavior.

These tasks span different coordination structures, ranging from loosely coupled navigation to tightly coupled physical control. It allows us to evaluate the strengths of different learning paradigms.

Experimental Setup

Our phase-1 study consists of **24 experiments**: 2 algorithms (**IPPO, MAPPO**) $\times$ 4 tasks (**VMAS Balance, VMAS Navigation, PettingZoo Multiwalker, PettingZoo Simple Tag**) $\times$ 3 seeds (**0, 1, 2**). All runs were launched through the BenchMARL framework. To keep the comparison fair and easy to interpret, each run used a training budget of **2,000,000 frames**, learning rate **3e-4**, entropy coefficient **0.01**, and the same underlying hardware. For these on-policy runs, BenchMARL collected **6000** frames per training iteration, resulting in roughly **334** total iterations per run. Evaluation was performed every **120,000** frames over **10 evaluation episodes**. Both policy and critic used MLP-based models with two hidden layers of size **256**. Overall, this setup provides a compact & diverse reproducibility study of PPO-based multi-agent baselines, covering both cooperative and role-asymmetric environments. We evaluate reproducibility using both quantitative and qualitative evidence.

Results & Analysis

Quantitatively, we summarize each method by its best evaluation reward, final evaluation reward, and the percentage drop from peak to final performance, aggregated across three seeds. Qualitatively, we inspect evaluation rollouts through

Task	Algo	Best	Final	Drop%
Navigation	MAPPO	1.17±0.04	0.97±0.08	16.8
Navigation	IPPO	1.14±0.02	0.87±0.19	23.7
Balance	MAPPO	54.06±2.38	45.37±2.78	16.1
Balance	IPPO	52.08±2.05	36.65±9.13	25.8
MWalker	MAPPO	5.98±2.59	3.02±1.14	49.4
MWalker	IPPO	9.74±0.92	-4.02±15.6	141.3
Tag	MAPPO	24.66±8.32	-5.33±43.04	–
Tag	IPPO	30.96±4.24	27.54±7.58	11.0

Table 1: *Aggregate evaluation summary across seeds. Best = peak mean reward, Final = last evaluation reward, Drop% = %age drop from peak to final.*

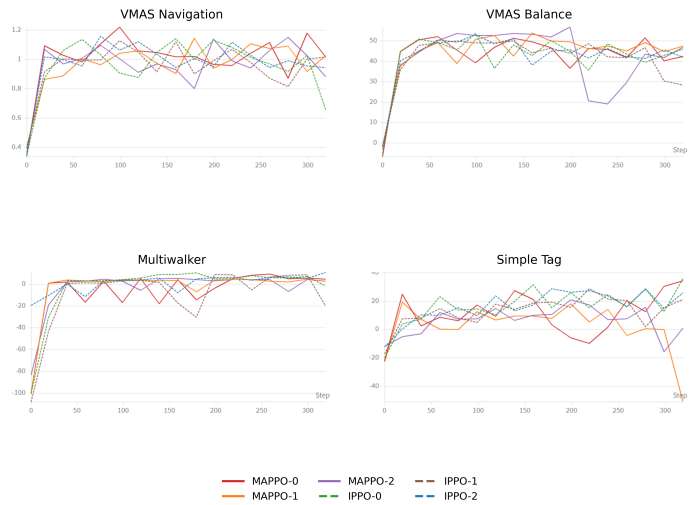

Figure 1: *Reward Curves for the 4 Environments*

representative early, mid, and late snapshots.

Navigation. Both algorithms learn effective navigation policies, but MAPPO is slightly stronger and more stable (Table 1, Fig. 1). MAPPO achieves better peak and final performance with a smaller drop, indicating more reliable training. Qualitatively, rollout snapshots show improved spatial separation and more direct trajectories over time, with reduced interference between agents in the late stage (Fig. 2). Overall, centralized value estimation provides a modest but consistent advantage in both stability and final performance.

Balance. Both algorithms achieve strong performance, but MAPPO is clearly more stable and attains better final results (Table 1, Fig. 1). While peak performance is comparable, MAPPO retains performance more effectively and exhibits lower variance across seeds. Qualitatively, successful runs show agents coordinating to stabilize and transport the shared object, whereas less stable policies fail to maintain consistent control over longer horizons (Fig. 2). Overall, Balance highlights a clear advantage of centralized value estimation in tightly coupled multi-agent control.

Multiwalker. Multiwalker shows the clearest trade-off between peak performance and stability (Table 1, Fig. 1). IPPO achieves higher peak rewards but suffers from severe degradation and high variance, often collapsing by the end of training. In contrast, MAPPO maintains lower but significantly more stable and positive final performance. Qualitatively, stable runs sustain coordinated locomotion and consistent package transport, whereas unstable ones collapse over longer horizons (Fig. 2). Overall, MAPPO provides more reliable and reproducible performance in this long-horizon coordination task.

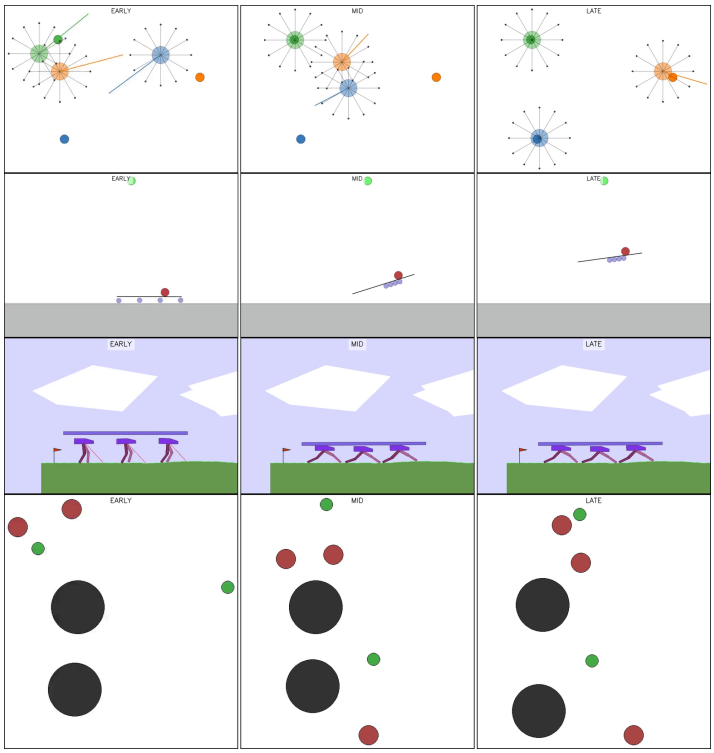

Figure 2: *Representative Rollouts with early, mid, and late stage snapshots. (From Top: Navigation, Balance, Multiwalker, Simple Tag)*

Simple Tag. In contrast to the cooperative tasks, Simple Tag exhibits substantially higher variability across seeds (Table 1, Fig. 1). Both methods are able to learn meaningful behaviors, and MAPPO achieves competitive peak performance. However, its training is less stable, with one run exhibiting a sharp late-stage collapse to a negative final reward. In comparison, IPPO remains more consistent across seeds with strong final performance. These results suggest that in competitive settings with misaligned objectives, centralized critics may be more sensitive to instability and initialization. We omit the drop percentage for MAPPO because one run collapses making the percentage less interpretable.

Reproducibility Takeaways. Across co-op tasks, MAPPO generally provides more stable training and stronger final performance than IPPO. However, the competitive Simple Tag setting reveals a limitation: although MAPPO can achieve strong intermediate performance, it is more sensitive to instability and can exhibit catastrophic degradation. Overall, these results suggest that centralized value estimation improves reproducibility in cooperative settings, but its advantages become less reliable when agent objectives are misaligned.